

Ch 3: Pair of Linear Equations in Two Variables

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Check graphically whether the pair $x + 3y = 6$ and $2x - 3y = 12$ is consistent. If yes, find the solution.

Q2. On comparing the ratios a_1/a_2 , b_1/b_2 , c_1/c_2 , find whether the pair $5x - 4y + 8 = 0$ and $7x + 6y - 9 = 0$ is consistent or inconsistent.

Q3. On comparing the ratios, find whether $9x + 3y + 12 = 0$ and $18x + 6y + 24 = 0$ is consistent, inconsistent, or dependent.

Q4. On comparing the ratios, find whether $6x - 3y + 10 = 0$ and $2x - y + 9 = 0$ is consistent or inconsistent. (PYQ 2020)

Q5. Find the value of k for which the pair $3x + 2y = 5$ and $2x - ky = 1$ has a unique solution.

Q6. Find the value of k for which the pair $kx + 2y = 5$ and $3x + y = 1$ has no solution.

Q7. Find the value of k for which the pair $2x + 3y = 7$ and $(k - 1)x + (k + 2)y = 3k$ has infinitely many solutions. (PYQ 2019)

Q8. Find the value of k for which the pair $kx - 5y = 2$ and $6x + 2y = 7$ has no solution.

Q9. For what value of k will the following pair of equations have infinitely many solutions: $2x + 3y = 4$ and $(k + 2)x + 6y = 3k + 2$. (PYQ 2018)

Q10. Find the value of k for which the system of equations $x + 2y = 3$ and $5x + ky = 15$ has infinitely many solutions.

Q11. Find the value of k for which the pair $4x + ky = 8$ and $2x + 2y = 2$ has a unique solution.

Q12. Find the value of k for which the pair $2x - ky = 7$ and $6x + 10y = 3$ represents parallel lines.

Q13. Solve by substitution: $3x - y = 3$ and $7x + 2y = 20$.

Q14. Solve by substitution: $x + y = 14$ and $x - y = 4$.

Q15. Solve by elimination: $3x + 4y = 10$ and $2x - 2y = 2$.

Q16. Solve by elimination: $2x + y = 6$ and $2x - y = 2$.

Q17. Solve: $x + y = 5$ and $2x - 3y = 5$.

Q18. Solve: $0.2x + 0.3y = 1.3$ and $0.4x + 0.5y = 2.3$.

Q19. Solve: $\sqrt{2}x + \sqrt{3}y = 0$ and $\sqrt{3}x - \sqrt{8}y = 0$.

Q20. Solve: $(3x/2) - (5y/3) = -2$ and $(x/3) + (y/2) = 13/6$. (PYQ type)

Q21. A system of two linear equations is inconsistent. What does this mean graphically?

Q22. A system has infinitely many solutions. What does this mean graphically? What is the condition on coefficients?

Q23. Write the condition for the pair $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to have: (i) a unique solution (ii) no solution (iii) infinitely many solutions.

Q24. The sum of a two-digit number and the number obtained by reversing the digits is 66. If the digits of the number differ by 2, find the number.

Q25. The difference between two numbers is 26 and one number is three times the other. Find them.

Q26. 5 years hence, the age of Sachin will be three times the age of his son. Find their present ages if the sum of their ages is 30.

Q27. A fraction becomes $\frac{9}{11}$ when 2 is added to both numerator and denominator. It becomes $\frac{5}{7}$ when 1 is subtracted from both. Find the fraction. (PYQ 2018)

Q28. Solve: $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$ (where $x \neq 0, y \neq 0$). (PYQ)

Q29. Solve: $\frac{1}{x+1} + \frac{2}{y-1} = \frac{5}{2}$ and $-\frac{3}{x+1} + \frac{1}{y-1} = 2$ for x and y .

Q30. If $2x + y = 23$ and $4x - y = 19$, find the values of $(5y - 2x)$ and $y/x - 2$. (PYQ 2020)

Q31. For what value of p will the following pair of equations have a unique solution: $px + 3y - (p-3) = 0$ and $12x + py - p = 0$. (PYQ)

Q32. Find k if the pair $(k-3)x + 3y = k$ and $kx + ky = 12$ has infinitely many solutions. (PYQ 2019)

Q33. Solve: $99x + 101y = 499$ and $101x + 99y = 501$. (PYQ 2022)

Q34. Solve: $217x + 131y = 913$ and $131x + 217y = 827$. (PYQ 2021)

SECTION B - 3 MARKS

Q1. Solve graphically: $x + 3y = 6$ and $3y - 2x = -12$. State whether consistent. (PYQ)

Q2. Solve graphically: $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Find the vertices of the triangle formed by the lines and the x-axis. (PYQ 2018)

Q3. Solve graphically: $2x + y - 6 = 0$ and $4x - 2y - 4 = 0$. Find the area of the triangle formed by the lines with the x-axis. (PYQ 2019)

Q4. Solve graphically: $x + 2y = 5$ and $2x - 3y = -4$. Also find the area of the triangle formed with the y-axis. (PYQ)

Q5. Draw the graph of $2x + y = 6$ and $2x - y + 2 = 0$. Find the vertices of the triangle formed by these lines and the x-axis. (PYQ 2020)

Q6. Draw the graph of $x + y = 3$ and $3x - 2y = 4$. Find the point of intersection and the area of triangle formed with the x-axis.

Q7. Solve by substitution method: $2x + 3y = 9$ and $4x + 6y = 18$. Comment on the nature of the solution.

Q8. Solve by substitution method: $7x - 15y = 2$ and $x + 2y = 3$.

Q9. Solve by substitution: $3x - 5y - 4 = 0$ and $9x = 2y + 7$. (PYQ 2018)

Q10. Solve by elimination: $3x + 4y = 10$ and $2x - 2y = 2$. (PYQ 2019)

Q11. Solve by elimination: $3x - 5y - 4 = 0$ and $9x = 2y + 7$.

Q12. Solve by elimination: $2/x + 3/y = 13$ and $5/x - 4/y = -2$. (PYQ)

Q13. Solve: $x/a + y/b = 2$ and $ax - by = a^2 - b^2$.

Q14. Solve: $(a-b)x + (a+b)y = a^2 - 2ab - b^2$ and $(a+b)(x+y) = a^2 + b^2$.

Q15. Solve: $ax + by = a - b$ and $bx - ay = a + b$. (PYQ 2018)

Q16. Solve: $(b - a)y - (a + b)x = 2b$ and $x + y = 0$ for x and y.

Q17. Solve: $x/3 + y/4 = 4$ and $5x/6 - y/8 = 4$.

Q18. Solve: $217x + 131y = 913$ and $131x + 217y = 827$. (PYQ 2021)

Q19. Solve: $99x + 101y = 499$ and $101x + 99y = 501$. (PYQ 2022)

Q20. Solve: $152x - 378y = -74$ and $-378x + 152y = -604$.

Q21. Solve by cross-multiplication: $2x + y = 5$ and $3x + 2y = 8$.

Q22. Solve by cross-multiplication: $x + y = a + b$ and $ax - by = a^2 - b^2$.

Q23. Solve by cross-multiplication: $x/a + y/b = a + b$ and $x/a^2 + y/b^2 = 2$. (PYQ)

Q24. Solve: $4/(x+y) + 3/(x-y) = 14$ and $3/(x+y) - 4/(x-y) = 23$. (PYQ)

Q25. Solve: $1/(3x+y) + 1/(3x-y) = 3/4$ and $1/2(3x+y) - 1/2(3x-y) = -1/8$. (PYQ)

Q26. Solve: $5/(x+y) - 2/(x-y) = -1$ and $15/(x+y) + 7/(x-y) = 10$. (PYQ 2019)

Q27. Solve: $10/(x+y) + 2/(x-y) = 4$ and $15/(x+y) - 9/(x-y) = -2$. (PYQ 2018)

Q28. Solve: $2/(x+2y) + 1/(2x-y) = 3$ and $5/(x+2y) - 3/(2x-y) = 1$.

Q29. The sum of a two-digit number and its reverse is 66. The digits differ by 2. Find all possible numbers. (PYQ 2019)

Q30. The sum of the digits of a two-digit number is 9. If 27 is added to the number, the digits get reversed. Find the number. (PYQ)

Q31. A two-digit number is 4 times the sum of its digits and twice the product of its digits. Find the number. (PYQ 2020)

Q32. The sum of the numerator and denominator of a fraction is 4 more than twice the numerator. If 3 is added to each, they are in the ratio 2:3. Find the fraction. (PYQ)

Q33. A fraction becomes $\frac{1}{3}$ when 1 is subtracted from numerator and it becomes $\frac{1}{4}$ when 8 is added to denominator. Find the fraction. (PYQ 2020)

Q34. 5 years ago, Nuri was thrice as old as Sonu. 10 years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu now? (PYQ 2019)

Q35. Two years ago, a father was 5 times as old as his son. 2 years later his age will be 8 more than 3 times the age of the son. Find their present ages. (PYQ)

Q36. The present age of a father is 3 years more than 3 times the age of his son. 3 years hence the father's age will be 10 years more than twice the son's age. Find their ages.

Q37. A train covered a certain distance at uniform speed. Had the train been 10 km/hr faster, it would have taken 2 hours less. Had it been 10 km/hr slower, it would have taken 3 more hours. Find the distance and speed. (PYQ 2018)

Q38. Ritu can row downstream 20 km in 2 hours and upstream 4 km in 2 hours. Find her speed of rowing in still water and the speed of current. (PYQ 2019)

Q39. A boat goes 30 km upstream and 44 km downstream in 10 hours. It goes 40 km upstream and 55 km downstream in 13 hours. Find the speed of the stream and the boat.

Q40. A person invested some amount at 12% per annum and some at 10% per annum. Total interest in one year is Rs. 130. If amounts are interchanged, interest becomes Rs. 4 less. Find the amounts.

Q41. 8 men and 12 boys can finish a piece of work in 10 days while 6 men and 8 boys can finish it in 14 days. Find the time taken by one man and one boy separately. *(PYQ)*

Q42. Find the value of k for which the system $(k+1)x + 8y = 4k$ and $kx + (k+3)y = 3k-1$ has infinitely many solutions. *(PYQ 2019)*

SECTION C - 5 MARKS

Q1. Solve graphically: $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Find the area of triangle formed by these two lines and the x -axis. *(PYQ 2018 — most repeated)*

Q2. Draw the graphs of $2x + y - 6 = 0$ and $4x - 2y - 4 = 0$. Determine the vertices of the triangle formed by these lines and the x -axis. Find the area. *(PYQ 2019)*

Q3. Draw the graphs of $2x - 3y + 6 = 0$ and $2x + 3y - 18 = 0$. Find the vertices of the triangle formed with the y -axis and find the area of the triangle. *(PYQ 2020)*

Q4. Solve graphically: $3x - y = 3$ and $2x - 3y = -7$. Shade the region between the two lines and the y-axis. Also find the area of this triangular region. (PYQ)

Q5. Solve graphically: $x + 2y = 5$ and $2x - 3y = -4$. Find the vertices of the triangle formed with the y-axis and find its area.

Q6. 5 prizes were distributed among students. Prize for hockey was x and for cricket was y . $5 \text{ hockey} + 4 \text{ cricket} = \text{Rs. } 9500$. $4 \text{ hockey} + 3 \text{ cricket} = \text{Rs. } 7370$. Find prize for each. (PYQ 2023 case-based)

Q7. Solve: $2/x + 3/y = 13$ and $5/x - 4/y = -2$. Hence find k if $k = y/x - 3$. (PYQ)

Q8. Solve: $4/(x+y) + 3/(x-y) = 14$ and $3/(x+y) - 4/(x-y) = 23$. (PYQ — most repeated)

Q9. Solve: $5/(x+y) - 2/(x-y) = -1$ and $15/(x+y) + 7/(x-y) = 10$. (PYQ 2019)

Q10. Solve: $10/(x+y) + 2/(x-y) = 4$ and $15/(x+y) - 9/(x-y) = -2$. (PYQ 2018)

Q11. Solve: $1/(2x) - 1/y = -1$ and $1/x + 1/(2y) = 8$ where $x \neq 0$, $y \neq 0$.

Q12. Solve: $(x+1)/(y+2) = 5/8$ and $(2x+1)/(y+5) = 2/5$ for x and y .

Q13. Solve $ax + by = 1$ and $bx + ay = (a+b)^2/(a^2+b^2) - 1$. (PYQ)

Q14. A boat covers 32 km upstream and 36 km downstream in 7 hours. It also covers 40 km upstream and 48 km downstream in 9 hours. Find the speed of the boat and the current. (PYQ 2019)

Q15. A train takes 3 hours less if its speed is increased by 12 km/hr. It would take 3 hours more if its speed is decreased by 12 km/hr. Find the original speed and the distance. (PYQ 2018)

Q16. Places A and B are 100 km apart. One car starts from A and another from B at the same time. If they travel in the same direction, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. Find the speed of both cars. (PYQ 2020)

Q17. A man travels 370 km partly by train and partly by car. If he covers 250 km by train and the rest by car, it takes 4 hours. But if he travels 130 km by train and the rest by car, it takes 18 minutes more. Find the speed of the train and car. (PYQ 2018)

Q18. Students of a class were made to stand in rows. If 4 students are extra in each row, there would be 2 rows fewer. If 4 students are fewer in each row, there would be 4 more rows. Find the number of students in the class. (PYQ 2020)

Q19. Meena went to a bank to withdraw Rs. 2000. She asked the cashier to give her Rs. 50 and Rs. 100 notes only. She got 25 notes in all. Find how many notes of each denomination she got. (PYQ)

Q20. Five years hence, the age of Jacob will be three times that of his son. Five years ago, Jacob's age was seven times that of his son. Find their present ages. *(PYQ 2019)*

Q21. The ages of two friends Ani and Biju differ by 3 years. Ani's father Dharam is twice as old as Ani and Biju is twice as old as his sister Cathy. The ages of Cathy and Dharam differ by 30 years. Find the ages of Ani and Biju. *(PYQ)*

Q22. Solve: $2(ax - by) + (a + 4b) = 0$ and $2(bx + ay) + (b - 4a) = 0$. *(PYQ)*

Q23. Solve: $(a+b)x + (a-b)y = a^2 - 2ab - b^2$ and $(a+b)(x+y) = a^2 + b^2$. *(PYQ HOTS)*

Q24. For what values of a and b will the following system of equations have infinitely many solutions: $2x + 3y = 7$, $(a-b)x + (a+b)y = 3a + b - 2$. *(PYQ 2019)*

Q25. Draw the graph of $x + y = 5$ and $x - y = 3$. Find area of triangle formed by the two lines and y -axis. Also determine whether the pair has a unique solution.